OAVS TGT · Current Affairs

Science, Space, Technology & Environment

Built from the science and environment questions in your 5 previous-year papers

Coverage window: Jan 2025 – 28 Oct 2026

How this part is organised

Science, space, technology and environment were **27 of 150** current-affairs questions (18%, or 4–8 per paper), tied with international affairs as the biggest share. **Air pollution and climate policy alone were 10**. The rest: technology and computing (6), biodiversity, disasters and reports (5), space (4), energy (2).

Section	Topic	Why it matters
1	Space: ISRO, Gaganyaan, private rockets	POEM-3, Skyroot, HysIS, lunar station asked
2	Technology: AI, semiconductors, digital law	5G, supercomputers, GARUDA grid, Geospatial hackathon
3	Climate and energy: COP30, NDC, renewables, Ramsar	Climate agenda, rooftop solar, Indian Railways net zero
4	Air pollution and Delhi policy	10 PYQs: CAQM, IQAir, NCAP, BS-VI, Winter Action Plan
5	Static quick-facts from your PYQs	HANSA-NG, Param Pravega, TX2, ICIMOD
6	Practice MCQs (25) and answer key	—

Reliability key. ✔ = checked against live web sources on 28 Oct 2026. ◆ = from my own knowledge, so cross-check once before the exam. Unmarked facts come from your PYQ keys.

What the PYQs tell us

Paper	Q	Topic asked	Type
Sci 2024	42	ISRO's first zero-debris-in-orbit mission (Mar 2024) – POEM-3	Space
Sci 2024	48	CAQM policy for Delhi-NCR (July 2022) – Policy to Curb Air Pollution	Air quality
Sci 2024	51	Delhi Jal Board: 3 areas to check water-polluting industries (Jan 2022)	Pollution
Sci 2024	53	Skyroot Aerospace – static test of Kalam-250 S2 motor (Mar 2024)	Space / startup
Sci 2024	62	5G services launched 1 Oct 2022	Technology
Sci 2024	65	SHOONYA – NITI Aayog campaign for EVs in ride-hailing and deliveries	Campaign
Sci 2024	66	IQAir World Air Quality Report 2021 – Delhi most polluted capital	Report
Sci 2024	68	Rooftop Solar Programme Phase-II – 40% subsidy for first 3 kW	Energy
Maths 2024	46	GARUDA – India's national grid computing initiative	Computing
Maths 2024	49	WHO Air Quality Database 2022 – 99% breathe air above limits	Report
Maths 2024	53	BS VI fuel standard adopted	Air quality
Maths 2024	70	HANSA-NG trainer aircraft by CSIR-NAL (May 2022)	Aviation
Eng 2019	3	Face of Disaster 2019 report – SEEDS	Report
Eng 2019	4	HysIS (Nov 2018) – Earth observation satellite	Space
Eng 2019	11	SWAS green crackers (CSIR)	Air quality
Eng 2019	18	AMR full form – Anti-Microbial Resistance	Health
Eng 2019	27	National Clean Air Programme – Environment Ministry	Air quality
Eng 2022	8	State of Global Air 2020 – India highest per-capita exposure	Report
Eng 2022	9	Param Pravega supercomputer at IISc Bengaluru (Feb 2022)	Computing
Eng 2022	16	China–Russia MoU on International Lunar Research Station (Mar 2021)	Space
Eng 2022	24	Cyclone Tauktae (May 2021)	Disaster
SS 2023	14	Delhi 15-point Winter Action Plan (Sept 2022)	Air quality
SS 2023	18	TX2 Award (Jan 2023) – Pench Tiger Reserve	Biodiversity
SS 2023	19	ICIMOD report on gender equality and climate change (Sept 2022)	Report
SS 2023	22	5G launched at the 6th India Mobile Congress	Technology
SS 2023	23	Indian Railways net zero carbon by 2030	Energy
SS 2023	27	Geospatial Hackathon (Jan 2023) – Dr Jitendra Singh	Technology

Patterns to exploit

- **Delhi air pollution returns every paper** (CAQM, Winter Action Plan, IQAir, SHOONYA, BS-VI). Section 4 has the 2026-27 plan and the latest IQAir ranking.
- **Space questions name a mission and ask what it did** (zero debris, Earth observation, static test). Learn the ISRO timeline in Section 1.
- **Technology questions ask 'what is it' or 'who built it'** (GARUDA, Param Pravega, HANSA-NG). Section 5 lists them.
- **Reports are asked by publisher and ranking** (IQAir, WHO, State of Global Air, ICIMOD).
- Questions came from the previous 12–18 months, so Gaganyaan, Vikram-1, the AI Impact Summit and the 2025 IQAir report are the top tier.

1. Space: ISRO, Gaganyaan and private rockets

Date	Event	Facts	Chk
14 Jan 2025	ISRO chairman	**Dr V. Narayanan** (propulsion expert from LPSC) took charge (earlier S. Somanath)	◆
Jan 2025	**SpaDeX docking**	Launched on PSLV-C60 (30 Dec 2024): SDX01 (Chaser) and SDX02 (Target), about 220 kg each. **First in-orbit docking confirmed 16 Jan 2025**; undocking completed in March 2025. A step towards Chandrayaan-4 and the space station	✔
29 Jan 2025	**ISRO's 100th launch**	GSLV-F15 placed navigation satellite **NVS-02** in orbit from Sriharikota	✔
18 May 2025	PSLV-C61 failure	EOS-09 (RISAT-1B) lost after a combustion-chamber pressure drop – PSLV's first failure after a long streak	✔
June–July 2025	**Axiom-4**	**Shubhanshu Shukla** became the first Indian astronaut on the International Space Station; 18 days on the ISS; returned July 2025	✔
30 July 2025	**NISAR**	NASA–ISRO Synthetic Aperture Radar (about $1.5 billion; L-band and S-band) launched on **GSLV-F16** into a 743 km sun-synchronous orbit; maps the whole globe every 12 days	✔
2 Nov 2025	**CMS-03**	LVM3-M5 launched the 4,410 kg communication satellite, the heaviest launched from India; first in-space restart of the C25 cryogenic engine	✔
Dec 2025	LVM3-M6	Launched the commercial BlueBird Block-2 satellite for AST SpaceMobile (largest commercial communication satellite in low orbit)	✔
12 Jan 2026	PSLV-C62 failure	EOS-N1 and 15 co-passengers lost (including DRDO's Anvesha) after a roll-rate deviation before third-stage separation – **second consecutive PSLV failure**	✔
10 Apr 2026	Gaganyaan test	Crew module of Gaganyaan-1 passed an air-drop (parachute) test	✔
18 July 2026	**Skyroot Vikram-1**	**India's first privately built orbital rocket reached orbit on its first flight** ('Mission Aagaman'; about 450 km orbit; 350 kg to LEO; four stages) from Sriharikota. India became the **third country with private orbital launch capability** (after USA and China). Skyroot became India's first space-tech unicorn (**$1.1 billion**, May 2026). Vikram-2 (1,000 kg) targeted for 2027	✔

Upcoming ISRO missions (status of 12 Oct 2026)

Mission	Latest status	Chk
Gaganyaan G1	First uncrewed test flight (carries the humanoid **Vyommitra**); ISRO says it is targeted **within 2026** but no date is announced. Two more uncrewed flights (G2, G3) precede the crewed mission H1, expected around 2027–28. Four astronaut-designates: **Shubhanshu Shukla, Prasanth Balakrishnan Nair, Ajit Krishnan, Angad Pratap**	✔
Bharatiya Antariksh Station	First module by **2028**; full station by 2035	✔
Chandrayaan-4	Moon sample-return mission, about 2028; Venus Orbiter Mission about March 2028	✔
NavIC	Seven-satellite constellation to be ready by 2027	✔
Your PYQ space links	**POEM-3** (PSLV orbital experiment module) – ISRO's zero-debris mission (Mar 2024) · **Skyroot Kalam-250** motor test (Mar 2024) · **HysIS** hyperspectral Earth-observation satellite (Nov 2018) · China–Russia **International Lunar Research Station** MoU (Mar 2021)	

2. Technology: AI, semiconductors and digital law

Topic	Facts	Chk
India AI Impact Summit 2026	**16–20 Feb 2026** (extended to 21 Feb), Bharat Mandapam, New Delhi – the **first AI summit hosted by a Global South nation**. Over 100 countries; expo with 300+ exhibitors from 30 countries. **New Delhi Declaration on AI Impact** adopted 19 Feb, endorsed by 89 countries and organisations at first (later 92), including the US and China; non-binding; seven 'chakras'. India formally joined **Pax Silica**, a US-led initiative on semiconductor supply chains. Investment pledges of $200–250 billion	✔
IndiaAI Mission	Sanctioned Mar 2024 at **₹10,371.92 crore**; about 34,000 GPUs deployed; 20,000 more announced; target **100,000 public GPUs by Dec 2026**; IndiaAI Mission 2.0 rolled out at the summit	✔
India Semiconductor Mission	10–12 projects approved (about ₹1.64 lakh crore): first commercial production in early 2026 (including Micron's plant for memory chips); **ISM 2.0 announced in Budget 2026-27**	✔
DPDP Rules 2025	Digital Personal Data Protection Rules notified **14 Nov 2025**; Data Protection Board set up; consent managers from about 13 Nov 2026; full compliance by **13 May 2027**	✔
UPI at 10	See the Finance part (24,162 crore transactions in FY26)	✔
Your PYQ technology items	**5G** launched 1 Oct 2022 at the 6th India Mobile Congress · **GARUDA** (C-DAC's national grid computing initiative) · **Param Pravega** (IISc Bengaluru, Feb 2022) · **HANSA-NG** (CSIR-NAL, May 2022 engine relight test) · **Geospatial Hackathon** (Jan 2023; Dr Jitendra Singh)	

Nobel Prizes in science 2025–26 (Physics, Chemistry, Medicine) are in the Awards part.

3. Climate, energy and biodiversity

Topic	Facts	Chk
COP30	**10–21 Nov 2025, Belém, Brazil** – the first COP in the Amazon; adopted the **Belém Package** (29 decisions, including a global adaptation indicator set and a push to scale climate finance to $1.3 trillion). COP31: Antalya, Türkiye (9–20 Nov 2026; Australia leads negotiations)	✔
India's NDC 3.0 (2035)	Submitted **24 Apr 2026**: **60% non-fossil electricity capacity by 2035** (conditional on international support), **47% emissions-intensity cut below 2005 levels**, carbon sink of 3.5–4 billion tonnes CO2e. The 2022 target was 50% non-fossil capacity and 45% intensity cut by 2030	✔
Renewable energy	India reached **50% non-fossil installed capacity in June–July 2025**, five years ahead of the 2030 goal. By 31 Mar 2026: 283.46 GW non-fossil (52.3% of about 533 GW). India has the world's third-largest renewable capacity (about 256 GW including large hydro in 2025-26). Forest and tree cover: 25.17% of land area	✔
Ramsar sites	**103 Ramsar sites** as of 15–16 Oct 2026, after **Sukhna Wetland (Chandigarh)** and **Sauj Jheel (Uttar Pradesh)** were added; third-highest in the world after the UK (176) and Mexico (144); Tamil Nadu has the most; India had 26 in 2014. Glaw Lake became Arunachal Pradesh's first (Aug 2026). Chilika (Odisha) and Keoladeo (1981) are the oldest	✔
World Heritage	**Maratha Military Landscapes of India** (12 forts in Maharashtra and Tamil Nadu) inscribed in 2025	✔
Rooftop solar	The PM Surya Ghar Muft Bijli Yojana (launched Feb 2024) supersedes the old Phase-II programme in your PYQ (40% subsidy for the first 3 kW); subsidy up to about ₹78,000 for 3 kW and above ◆	◆
Your PYQ items	TX2 Award (Jan 2023): **Pench Tiger Reserve** · Cyclone Tauktae (May 2021) hit Gujarat, Kerala, Karnataka and Goa · ICIMOD (Sept 2022) report on gender equality and climate change in South Asia and the HKH region · Indian Railways net zero by 2030	

4. Air pollution and Delhi policy

Topic	Facts	Chk
IQAir World Air Quality Report 2025	Released 24 Mar 2026 (8th edition; 143 countries, 9,446 cities). **India ranked 6th most polluted country** (PM2.5 48.9 µg/m³, about 10 times the WHO limit of 5). Top five: **Pakistan** (67.3), Bangladesh, Tajikistan, Chad, DR Congo. **Loni (Uttar Pradesh) was the world's most polluted city** (112.5 µg/m³); Byrnihat (Meghalaya) 3rd; Delhi 4th; Ghaziabad 5th. **New Delhi is the world's most polluted capital** (82.2 µg/m³, a three-year low). 130 of 143 countries exceeded the WHO guideline; only 13 met it	✔
Delhi Winter Action Plan 2026-27	Winter Air Quality Management Framework notified **19 June 2026** (under the Environment (Protection) Act, 1986). Delhi CM **Rekha Gupta** announced a 10-point plan: **'No PUCC, No Fuel'** at all 500 ANPR-equipped petrol pumps from 1 Oct; **ban on dust-generating demolition and outdoor civil construction 1 Nov–31 Jan**; AI-enabled Command and Control Centre with 334 enforcement teams; higher parking charges, staggered office timings and work-from-home in the winter protocol; EV Policy 2026 and Parivartan Yojana	✔
GRAP	Graded Response Action Plan, run by **CAQM** in Delhi-NCR since 2020: Stage I Poor (AQI 201–300), II Very Poor (301–400), III Severe (401–450), IV Severe+ (above 450). CAQM revised GRAP on 28 Sept 2026	✔
Static anchors from your PYQs	**NCAP** (National Clean Air Programme) – Environment Ministry (2019) · **BS-VI** fuel standard (from April 2020) · **CAQM** (Commission for Air Quality Management, 2021) adopted the 'Policy to Curb Air Pollution' (July 2022) · **SHOONYA** (NITI Aayog; EVs in ride-hailing and deliveries) · **SWAS** green crackers (CSIR, 2018) · WHO Air Quality Database 2022: 99% breathe polluted air · State of Global Air 2020: India highest exposure	

Trap: 'most polluted capital' is New Delhi, but the 'most polluted city' in the 2025 report is Loni, and the 'most polluted country' is Pakistan. India is 6th.

5. Static quick-facts from your PYQs

Item	Facts
POEM-3	PSLV Orbital Experimental Module-3: used the spent PSLV fourth stage as a platform and de-orbited it, ISRO's zero-debris-in-orbit mission (Mar 2024)
Kalam-250	Solid rocket motor from Skyroot (named after A. P. J. Abdul Kalam); the S2 stage static test was supported by ISRO (Mar 2024). Skyroot's Kalam-1200 test followed in Aug 2025
HysIS	Hyperspectral Imaging Satellite for Earth observation, launched Nov 2018 on PSLV-C43
Param Pravega	Supercomputer at IISc Bengaluru (about 3.3 petaflops), under the National Supercomputing Mission
GARUDA	Global Access to Resources Using Digital Architecture – the national grid computing initiative by C-DAC ◆
HANSA-NG	Two-seater trainer by CSIR-National Aerospace Laboratories (NAL)
AMR	Anti-Microbial Resistance (microbes resisting antibiotics and similar drugs)
Face of Disaster 2019	Report by SEEDS (Sustainable Environment and Ecological Development Society)

6. Practice MCQs (25)

Answers are on the last page. Each question uses facts marked ✔ above.

1. The names of the SpaDeX satellites:
(A) SDX01 Chaser and SDX02 Target (B) Aditya and Surya (C) Shakti and Garuda (D) NavIC-1 and NavIC-2

2. ISRO's 100th launch from Sriharikota carried:
(A) CMS-03 (B) NVS-02 (C) NISAR (D) EOS-09

3. PSLV-C61 (May 2025) carried which satellite?
(A) EOS-N1 (B) BlueBird (C) EOS-09 (D) XPoSat

4. NISAR was launched on:
(A) PSLV-C62 (B) LVM3-M5 (C) GSLV-F15 (D) GSLV-F16

5. The first Indian astronaut on the ISS (Axiom-4):
(A) Shubhanshu Shukla (B) Rakesh Sharma (C) Prasanth Nair (D) Angad Pratap

6. The heaviest communication satellite launched from India (4,410 kg):
(A) GSAT-11 (B) CMS-03 (C) INSAT-3DS (D) BlueBird-2

7. PSLV-C62 (12 Jan 2026) was:
(A) Successful (B) A partial success (C) A failure, the second in a row (D) Postponed

8. India's first privately built orbital rocket:
(A) Agnibaan (B) Vikram-S (C) RHUMI (D) Vikram-1

9. Vikram-1's maiden mission is named:
(A) Aagaman (B) Prarambh (C) Pragati (D) Udaan

10. Skyroot became India's first space-tech unicorn at a valuation of:
(A) $500 million (B) $1.1 billion (C) $2 billion (D) $250 million

11. Which humanoid will fly on Gaganyaan G1?
(A) Gaganmitra (B) Rudra (C) Vyommitra (D) Anvesha

12. The Bharatiya Antariksh Station's first module is planned by:
(A) 2026 (B) 2030 (C) 2035 (D) 2028

13. The India AI Impact Summit was held in:
(A) 16–20 Feb 2026 (B) 10–14 Jan 2026 (C) 1–3 Mar 2026 (D) 5–8 Dec 2025

14. The AI summit's non-binding outcome document:
(A) Bengaluru Accord (B) New Delhi Declaration on AI Impact (C) Delhi Protocol (D) Bharat AI Charter

15. India formally joined which US-led initiative at the AI summit?
(A) Quad AI (B) Chip Alliance (C) Pax Silica (D) Clean Network

16. IndiaAI Mission outlay:
(A) ₹5,000 crore (B) ₹7,200 crore (C) ₹12,000 crore (D) ₹10,371.92 crore

17. The DPDP Rules 2025 were notified on:
(A) 14 Nov 2025 (B) 1 Jan 2026 (C) 26 Jan 2026 (D) 15 Aug 2025

18. India submitted its 2035 NDC on:
(A) 10 Nov 2025 (B) 24 Apr 2026 (C) 5 Jun 2026 (D) 1 Jan 2026

19. India's 2035 NDC non-fossil capacity target:
(A) 50% (B) 55% (C) 60% (D) 70%

20. COP30 was hosted at:
(A) Baku (B) Dubai (C) Glasgow (D) Belém

21. India crossed 50% non-fossil installed capacity in:
(A) 2025 (B) 2027 (C) 2029 (D) 2023

22. India's Ramsar sites as of Oct 2026:
(A) 96 (B) 103 (C) 91 (D) 110

23. Which was named the world's most polluted city in the 2025 IQAir report?
(A) Delhi (B) Byrnihat (C) Loni (D) Lahore

24. India's rank among most polluted countries in IQAir 2025:
(A) 3rd (B) 4th (C) 5th (D) 6th

25. Which state has the most Ramsar sites?
(A) Tamil Nadu (B) Odisha (C) Kerala (D) Gujarat

Answer key

1-A **2**-B **3**-C **4**-D **5**-A **6**-B **7**-C **8**-D **9**-A **10**-B **11**-C **12**-D **13**-A **14**-B **15**-C **16**-D **17**-A **18**-B **19**-C **20**-D **21**-A **22**-B **23**-C **24**-D **25**-A

Before you rely on this: check whether Gaganyaan G1 has launched by your exam date, and the ◆ items (ISRO chairman date, PM Surya Ghar subsidy). I did not find the 2026 tiger census or cheetah updates. Reply **'next'** for the next part: Government schemes, policies and miscellaneous (days, culture, books).